Extracting Implied Constraints from BTOR2 Models

Btor2Sieve — Mid-Project Presentation

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Project Goal

- **Goal:** Automatically extract hidden **word-level** constraints C over bit-vectors and arrays implied by safety property P
- **Input:** Word-level **BTOR2** hardware model and safety property $P = \neg\text{bad}$
- **Output:**
  - Independently SMT-validated constraints ($P \Rightarrow C$)
  - Summary reports and standalone `.smt2` obligations
  - **Pono** verification results (`verify.json`, `verify.csv`)

Benchmarks & Preprocessing

Group	Dataset	Evaluation Purpose
1. **Hand-Crafted**	`examples/*.btor2`	Fast regression tests with known results
2. **Bit-Vector Track**	`HWMCC'25 (bv)`	Bounds, equalities, implications, bit-masks
3. **Array Track**	`HWMCC'25 (arrays)`	Native SMT `select`/`store` & array relations
4. **Multi-Property**	`CAV'18` (*PicoRV32, ZipCPU*)	Cross-property constraint reuse (`--reuse`)

Why Add `CAV'18` Benchmarks?

HWMCC splits designs into single-property files (1 `bad` each). `CAV'18` models contain up to 57 `bad` properties per design for testing `--reuse`.

Benchmark Preprocessing

Splits multi-property `CAV'18` models into individual single-property files (1 `bad` each) while preserving original line and property IDs for direct comparison.

Workflow



1. **Parser P via Pono: Pono** is the BTOR2 frontend; it reads the model and prints the formula.
2. **Subterm Vocabulary:** Extract words, arrays, constants, and predicates directly from P's subterms.
3. **Propose Candidates:** Generate untrusted candidate formulas C (*bounds, eq/diseq, implications, masks, arrays*) via template rules and solver search.
4. **Validate & Replay:** Prove $P \Rightarrow C$ in a Z3 solver, and replay exported `.smt2` queries in **cvc5**.
5. **Verify with Pono:** Write accepted C back as `constraint` and measure model checking in **Pono**.

Experimental Results

Overall Summary (413 Benchmarks)

- **Models: 373** completed ($\leq$ 46 s each), 40 hit 300s CPU limit, 0 errors
- **Properties: 543** analyzed with 100% complete search (0 `UNKNOWN`)
- **Candidates: 7,409 accepted** / 122,660 proposed (**6.0%**), 0 vacuous
- **Cross-Property Reuse: 689** accepted across multi-property models
- **SMT Replay: 8,641** `.smt2` obligations checked in **Z3** & **cvc5** with **0 mismatches**

Accepted by Family

Family	Accepted	Candidates
Implication	3,321	69,727
Bounds (Signed/Unsigned)	3,008	26,428
Disequality / Equality	1,070	24,724
Array / Masked bits	10	1,781

Accepted by Benchmark Track

Track	Props	Accepted	Rate
`CAV'18 (multi)`	178	859	10.8%
`CAV'18 (single)`	180	857	10.8%
`HWMCC'25 (arrays)`	80	1,833	8.7%
`HWMCC'25 (bv)`	105	3,860	4.5%

Mid-Project Status & Next Steps

Completed

- **Pono** BTOR2 frontend (`pono.py`) + subterm vocabulary extraction
- Template + solver-guided generation across all 6 constraint families
- Independent Z3 validator, **cvc5** `.smt2` replay, and `btor2sieve verify`
- Added `HWMCC'25` + `CAV'18` benchmarks and Pono-checked conversion/splitting scripts

Next Steps

- Define the corresponding implication obligation for candidates derived directly from the model
- Analyze per-family acceptance rates, extraction/validation times, and cross-property reuse